EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 04

Vectors

Components, Dot Product and Practice

By - Excellence Programme

3 October 2026

Topics to be covered

01 Adding by components

02 Change in velocity and rotation

03 Dot product

04 Unit vector practice

05 Angle and direction from components

06 Fresh JEE-style practice

07 Cross product (added)

What we are skipping today

Already done in Vector 01 and Vector 02 before slide 91:

· scalar multiply, compass language, reverse/scale angles

· rectangular components and magnitude

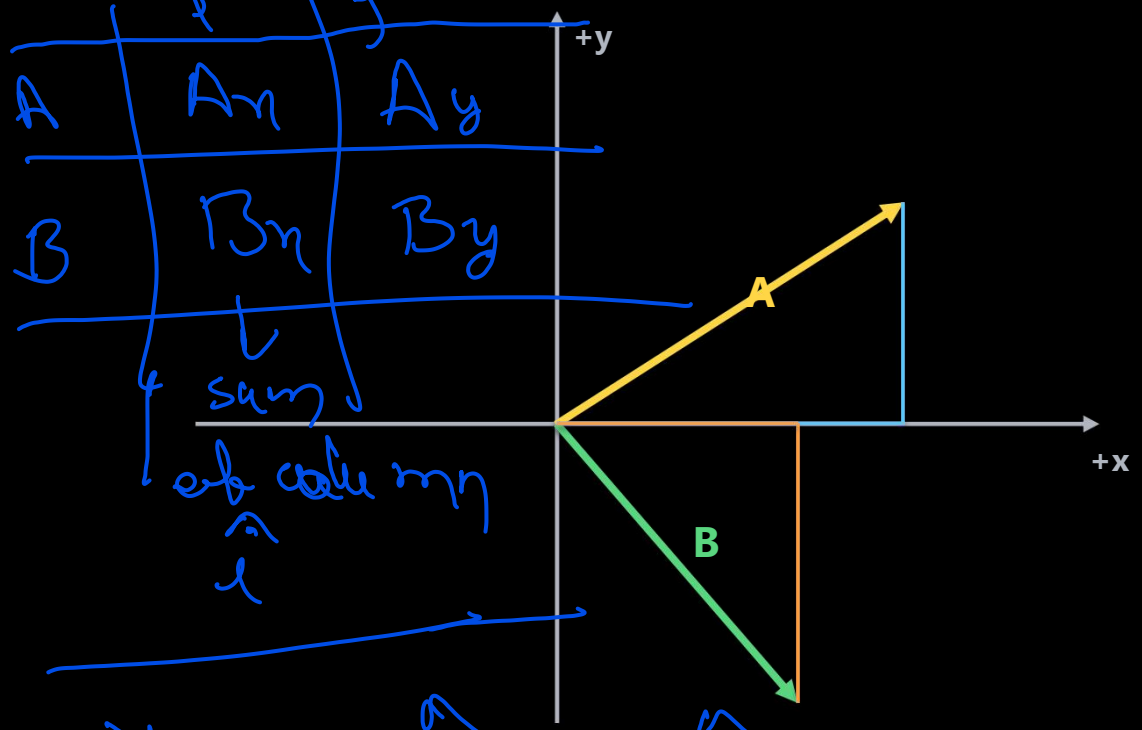
· triangle law, resultant formula, range, equal-magnitude tricks

· subtraction idea, and the repeated resultant drills

Today starts at adding by components, then moves into dot product and new practice. Cross product is added at the end.

Adding by components is adding columns

The x-parts combine with x-parts. The y-parts combine with y-parts. This is the cleanest method for many vectors.



$$\mathbf{A + B = (A_x + B_x)\,\hat{i} + (A_y + B_y)\,\hat{j}}$$

Column idea

x-column decides left-right. y-column decides up-down.

3D same rule

$$A + B = (A_x+B_x)\hat{i} + (A_y+B_y)\hat{j} + (A_z+B_z)\hat{k}$$

Add by components

Solve 1b

A = 4 î + 3 ĵ and B = 3 î - 4 ĵ. Find A + B, A - B, and |A| + |B|.

$A + B = 7\hat{i} + \hat{j}$

$A - B = \hat{i} + 7\hat{j}$

$|A| + |B| = 10$

Good

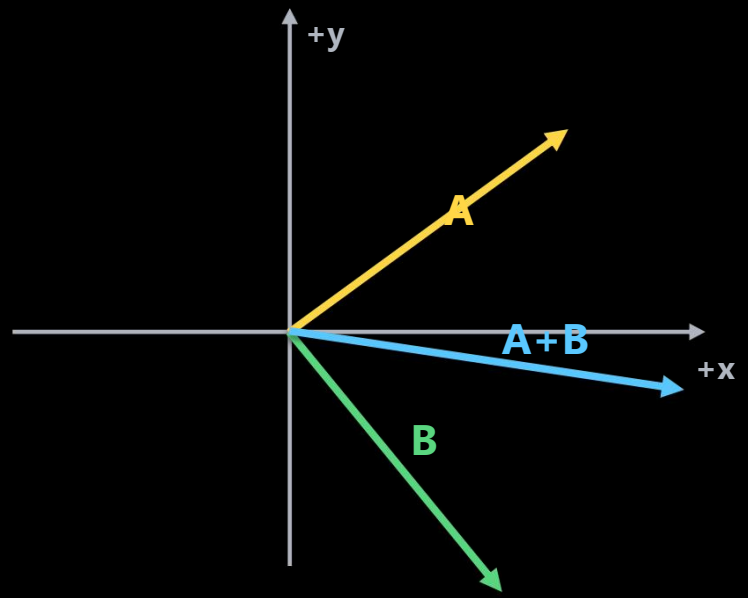
Solution: Add by components

Step 1: $\vec{A} + \vec{B} = 7\,\hat{i} - 1\,\hat{j}$

Step 2: $\vec{A} - \vec{B} = 1\,\hat{i} + 7\,\hat{j}$

Step 3: $|\vec{A}| = 5$ and $|\vec{B}| = 5$, so $|\vec{A}| + |\vec{B}| = 10$

Sum of magnitudes is 10. Magnitude of the sum is √50.

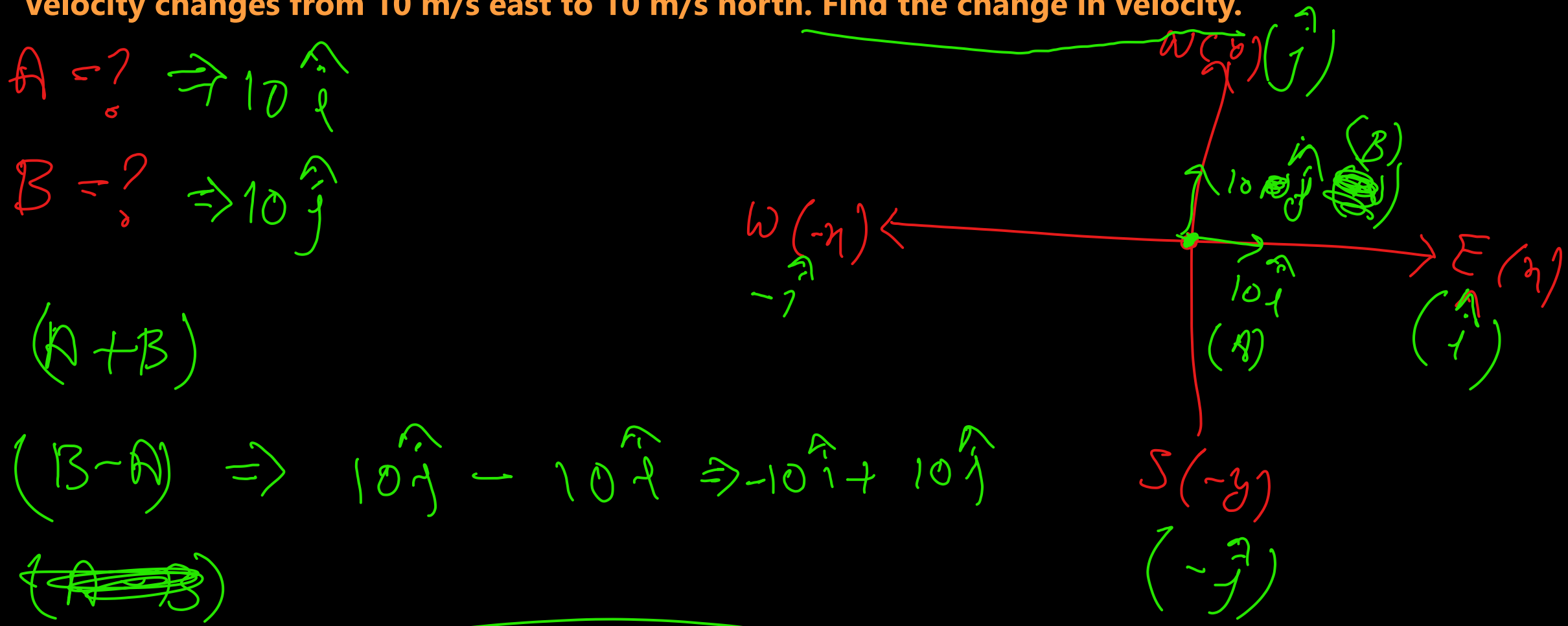


Rough work / extra work

Change in velocity

Solve It

Velocity changes from 10 m/s east to 10 m/s north. Find the change in velocity.

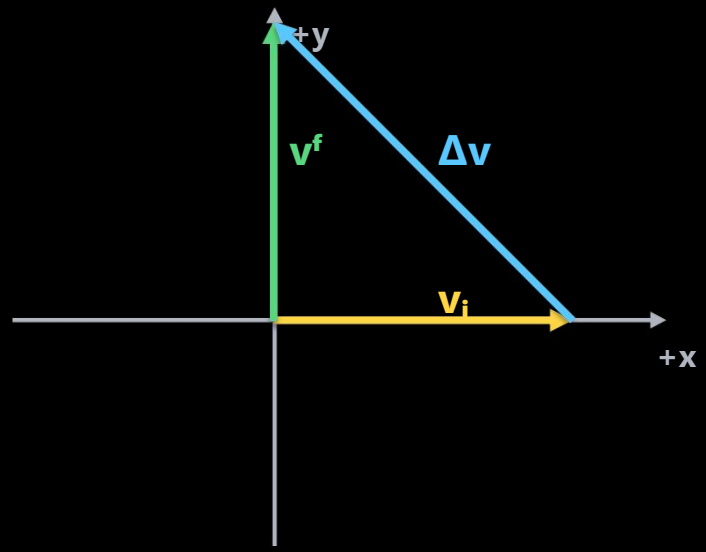


Solution: Change in velocity

Step 1: Change = final - initial = 10 ĵ - 10 î

Step 2: Magnitude = √(100 + 100) = 10 √2

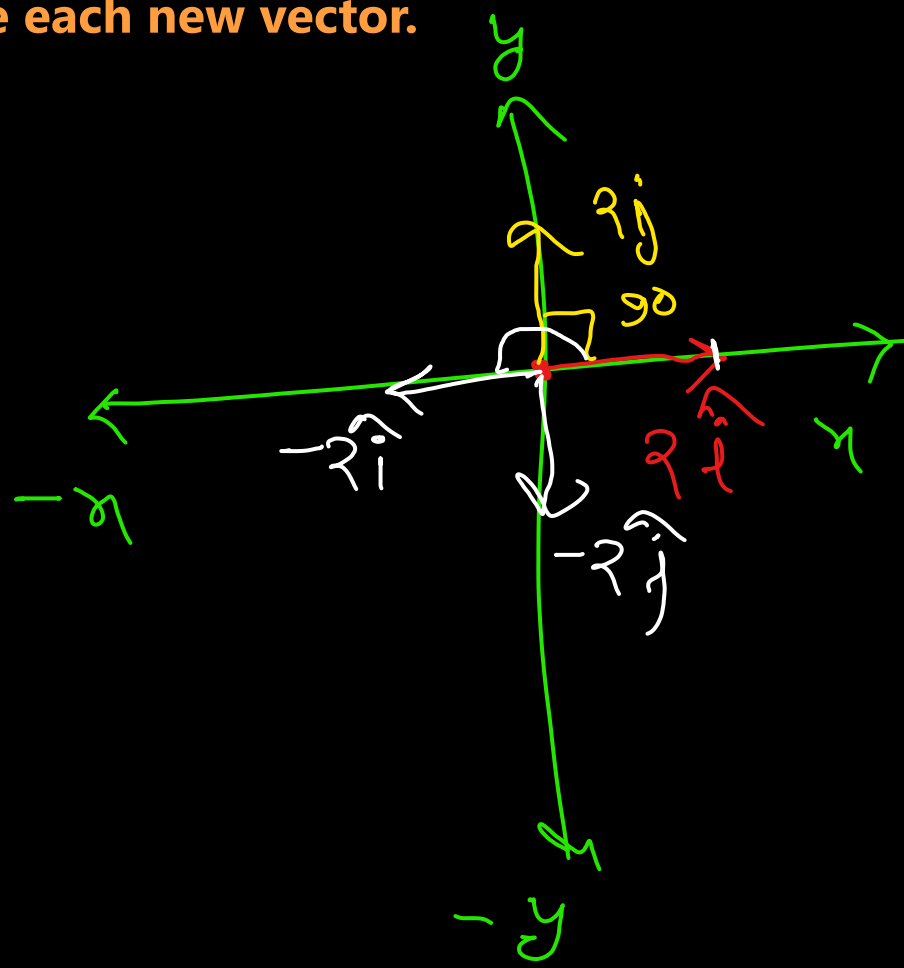
10 √2 m/s, pointing north of west.



Rotate 2 î

Solve It

A = 2 î is rotated by 90°, 180°, 270° and 360° anticlockwise. Write each new vector.



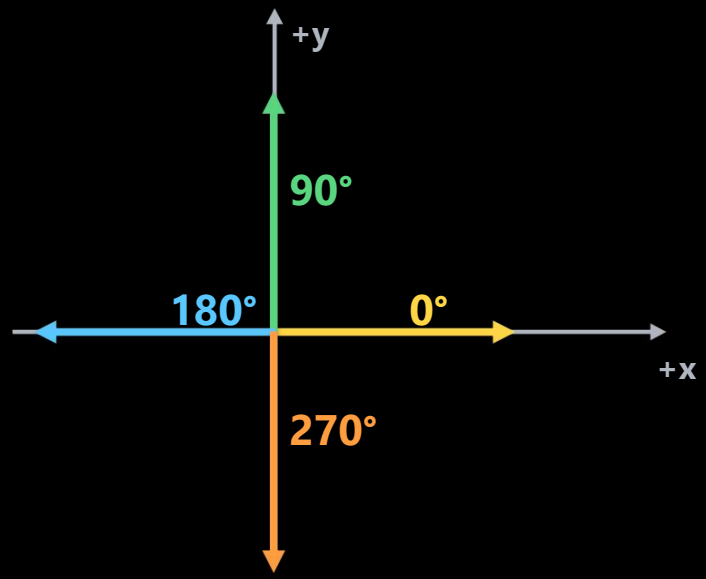
Solution: Rotate 2 î

Step 1: 90°: 2 ĵ

Step 2: 180°: -2 î

Step 3: 270°: -2 ĵ. 360°: 2 î again

Rotation keeps the magnitude and changes only the direction.



Rough work / extra work

Unit vector Along x-axis = $\hat{i}$

Unit vector Along y-axis = $\hat{j}$

z-axis = $\hat{k}$

$\hat{i} \cdot \hat{j} = 1 \times 1 \times \cos\theta \Rightarrow \cos 90 \Rightarrow 0$

($|\hat{i}| = 1$)

$A \cdot B = |A| \cdot |B| \cos\theta$

$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{i} \cdot \hat{k} = 0$

Dot product

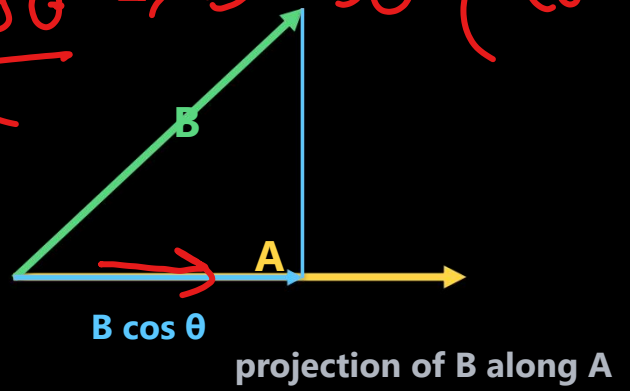
$AB\cos\theta = 0 \Rightarrow \cos\theta = \theta \Rightarrow \theta = 90$

$$\mathbf{A \cdot B = A\,B\cos\theta = A_x B_x + A_y B_y + A_z\,B_z}$$

If the dot product is 0, the vectors are perpendicular.

$\cos\theta = (A \cdot B) / (A\,B)$. That is how the angle is found from components.

Component of B along A = $(A \cdot B) / A$. $\Rightarrow \frac{A \cdot B}{A} \Rightarrow \frac{AB\cos\theta}{A} \Rightarrow B\cos\theta$ (comp along A)



$\hat{i} \cdot \hat{i} = 1$. $\hat{i} \cdot \hat{j} = 0$. The same rule holds for $\hat{j}$ and $\hat{k}$.

A unit vector dotted with itself is 1.

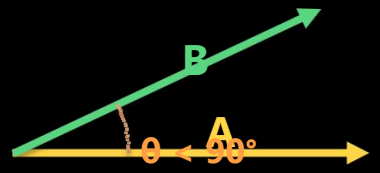
$A \cdot B =$

$A \times B =$

$$\cos\theta = \frac{\vec{A} \cdot \vec{B}}{|A| \, |B|}$$

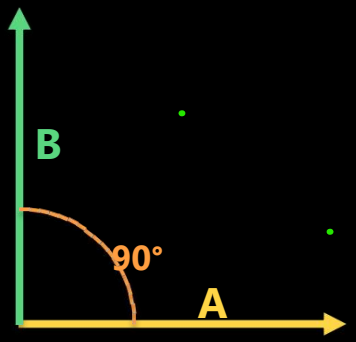
Dot product tells alignment

Dot product is positive when vectors face mostly the same way, zero at 90°, and negative when they face opposite way.



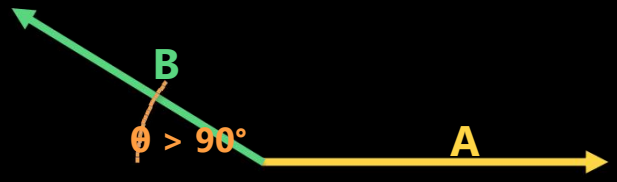
A · B > 0

$\cos(0) > 0$



A · B = 0

$\cos\theta = 0$



A · B < 0

$\cos(\theta) < 0$

Why this matters

It quickly tests perpendicular vectors and helps find the angle between two vectors.

Are these two vectors perpendicular?

Solve It

A = 3 î + 4 ĵ + 5 k̂ and B = 3 î + 4 ĵ - 5 k̂. What is the angle between them?

$\theta = ?^\circ \Rightarrow \theta = 90$

$\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$

$\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$

$$\cos\theta = \frac{\vec{A}\cdot\vec{B}}{|A||B|}$$

$$= \frac{A_x B_x + A_y B_y + A_z B_z}{|A||B|}$$

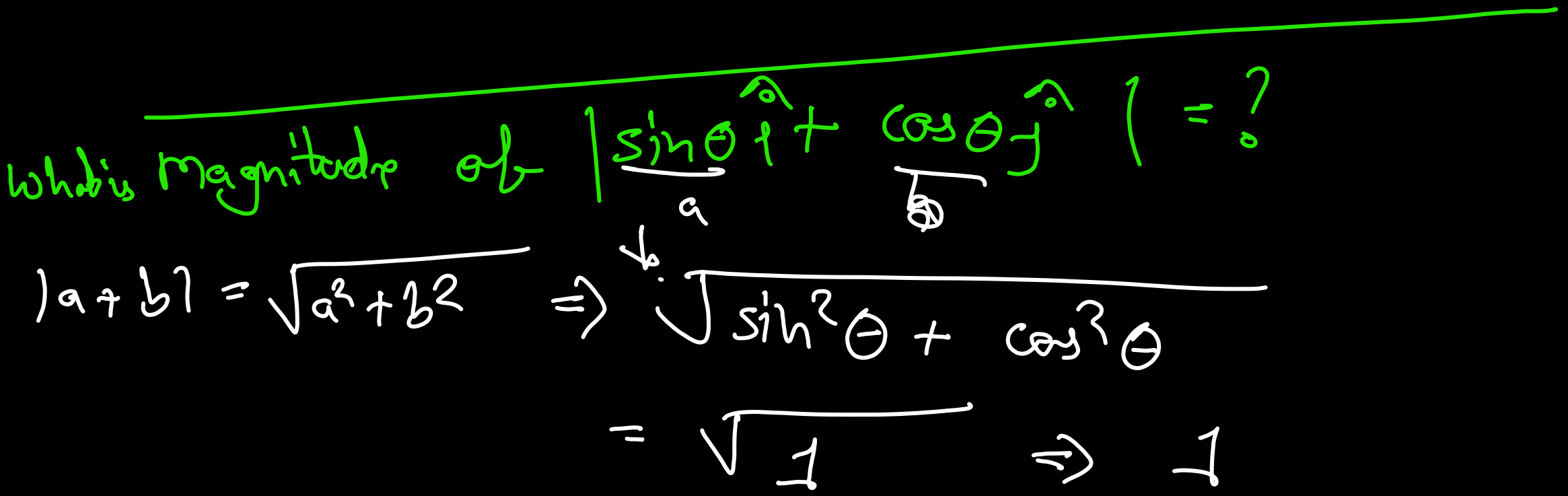
$$\frac{3\times 3 + 4\times 4 - 5\times 5}{|A||B|} = 0$$

$\cos\theta = 0 \Rightarrow \theta = 90^\circ$

Solution: Are these two vectors perpendicular?

Step 1: A · B = 9 + 16 - 25 = 0

The angle is 90°.



Angle with 4 î

Solve It

A = 2 î + 6 ĵ + 3 k̂ and B = 4 î. Find cos θ.

$$\cos\theta = \frac{A \cdot B}{|A||B|}$$

$$A \cdot B = 2\times 4 + 6\times 0 + 3\times 0$$

$$= 8$$

$$\Rightarrow \frac{8}{\sqrt{2^2+6^2+3^2}\times\sqrt{4^2}}$$

$$\Rightarrow \frac{8}{\sqrt{49}\times\sqrt{4^2}}$$

$$\Rightarrow \frac{8}{7\times 4}$$

$$\Rightarrow \frac{2}{7}$$

Solution: Angle with 4 î

Step 1: $A \cdot B = 8$

Step 2: $|A| = \sqrt{(4 + 36 + 9)} = 7$ and $|B| = 4$

Step 3: $\cos\theta = 8 / 28 = 2/7$

$\theta = \cos^{-1}$ of $2/7$.

$$\cos\theta = \frac{2}{7}$$

$$\theta = \cos^{-1}\left(\frac{2}{7}\right)$$

$$\cos\theta = \cos\left(\cos^{-1}\left(\frac{2}{7}\right)\right)$$

$$= \frac{2}{7}$$

Rough work / extra work

we want unit vector $\Rightarrow |A| = 1$

If we have $|A| = 5$

unit vector of $\hat{i} \Rightarrow$

unit vector of $\hat{i} + \hat{j} =$

$|\hat{i} + \hat{j}| \Rightarrow \sqrt{1^2 + 1^2} \Rightarrow \sqrt{2}$

$\Rightarrow \frac{A}{|A|}$

now $|\hat{A}| = 1$

$$\hat{A} = \frac{A}{|A|} \Rightarrow \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

$$\hat{A} = \frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}}$$ (unit vector of vector A)

Unit vectors from components

$\hat{i}$ itself is already a unit vector. → $\hat{i} \Rightarrow |1\hat{i}| = 1$ (Already 1)

Unit vector of $\hat{i} + \hat{j}$ is $(\hat{i} + \hat{j}) / \sqrt{2}$. → $|1\hat{i} + \hat{j}| = \sqrt{2}$

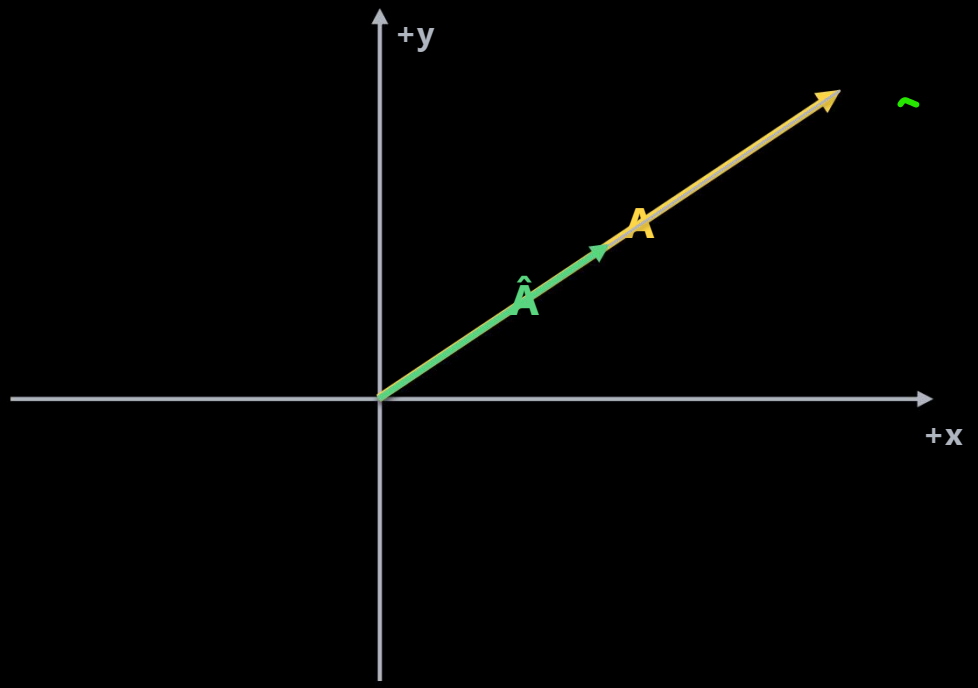
Unit vector of $\hat{i} + \hat{j} - \hat{k}$ is $(\hat{i} + \hat{j} - \hat{k}) / \sqrt{3}$. → $|1\hat{i} + \hat{j} - \hat{k}| = \sqrt{1^2 + 1^2 + (-1)^2} = \sqrt{3}$

$\sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ already has magnitude 1, because $\sin^2 + \cos^2 = 1$.

Unit vector of $3\hat{i} + 4\hat{j}$ is $(3\hat{i} + 4\hat{j}) / 5$.

Unit vector keeps direction and removes size

A unit vector is only the direction of a vector. It has length 1 and no physical unit.



$$\hat{A} = A / |A|$$

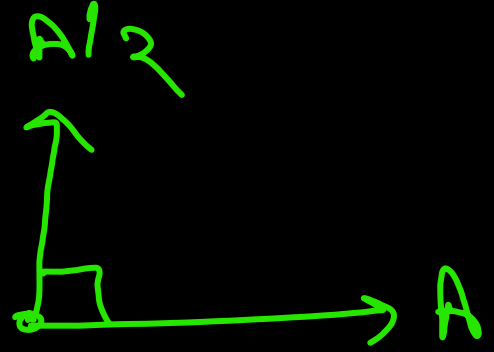
Example

For $A = 3\hat{i} + 4\hat{j}$, $|A| = 5$, so $\hat{A} = (3\hat{i} + 4\hat{j})/5$.

Do not attach N, m/s or kg to a unit vector.

Perpendicular forces A and A/2

Two forces of magnitudes A and A/2 are perpendicular. Find the magnitude of their resultant.



$\theta = 90^\circ$

$\vec{R} = \vec{A} + \vec{A/2}$

$|R| = \sqrt{A^2 + (A/2)^2 + 2A \cdot \frac{A}{2} \cos\theta}$

$\Rightarrow \sqrt{A^2 + \frac{A^2}{4} + \frac{2A^2}{2} \cos(90^\circ)}$ ($\cos(90^\circ) = 0$)

$\Rightarrow \sqrt{\frac{A^2}{1} + \frac{A^2}{4}} \Rightarrow \sqrt{\frac{5A^2}{4}} \Rightarrow \frac{\sqrt{5}A}{2}$

Solution: Perpendicular forces A and A/2

Step 1: cos 90° = 0, so $R = \sqrt{(A^2 + A^2/4)}$

Step 2: $R = \sqrt{(5\,A^2 / 4)} = (A/2)\,\sqrt{5}$

$(\sqrt{5} / 2) \times A$.

$\frac{\sqrt{5}}{2} A$

Rough work / extra work

Acceleration of a 4 kg body

$F_1 = 5\,\hat{i} + 8\,\hat{j} + 7\,\hat{k}$ and $F_2 = 3\,\hat{i} - 4\,\hat{j} - 3\,\hat{k}$ act on 4 kg. Find the acceleration.

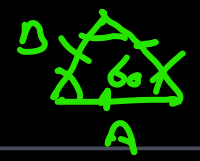
Solution: Acceleration of a 4 kg body

Step 1: Net force = $8\,\hat{i} + 4\,\hat{j} + 4\,\hat{k}$

Step 2: a = net force / 4

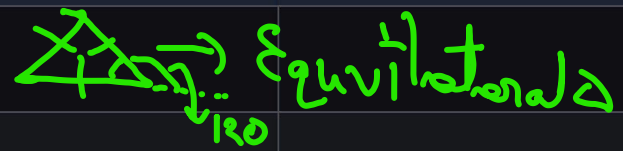
$a = 2\,\hat{i} + 1\,\hat{j} + 1\,\hat{k}$.

Angle conditions to lock





Condition	Angle
$\|A\| = \|B\| = \|A + B\|$	120°
$\|A\| = \|B\| = \|A - B\|$	60°
$\|A + B\| = \|A - B\|$	90°
$\|A + B\| = \|A\| + \|B\|$	0°
$\|A - B\| = \|A\| + \|B\|$	180°
$\|A + B\| = \|A\| - \|B\|$	180°
$\|A - B\| = \|A\| - \|B\|$	0°



Rough work / extra work

$A = 6\,\hat{i}$ and $B = 8\,\hat{j}$

H.W

Find $A + B$, $A - B$, $|A| + |B|$, and $|A + B|$.

Solution: A = 6 î and B = 8 ĵ

Step 1: A + B = 6 î + 8 ĵ

Step 2: A - B = 6 î - 8 ĵ

Step 3: |A| + |B| = 6 + 8 = 14, and |A + B| = √(36 + 64) = 10

Sum of lengths is 14. Length of the sum is 10.

Unit vector list

$\hat{i}$ itself is already a unit vector.

Unit vector of $\hat{i} + \hat{j}$ is $(\hat{i} + \hat{j}) / \sqrt{2}$.

Unit vector of $\hat{i} + \hat{j} - \hat{k}$ is $(\hat{i} + \hat{j} - \hat{k}) / \sqrt{3}$.

$\sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ already has magnitude 1.

Unit vector of $3\hat{i} + 4\hat{j}$ is $(3\hat{i} + 4\hat{j}) / 5$.

$$\hat{A} = A / |A|$$

Rough work / extra work

Find β for a unit vector

A = 0.4 î + 0.3 ĵ + β k̂ is a unit vector. Find β.

$|A| = 1$

$0.4^2 + 0.3^2 + B^2 = 1$

$B^2 = 0.75$

$0.16 + 0.09 + B^2 = 1$

$B^2 = 1 - 0.25$

$= 0.75$

Solution: Find β for a unit vector

Step 1: For a unit vector, $|A| = 1$

Step 2: $\sqrt{(0.16 + 0.09 + \beta^2)} = 1$

Step 3: $0.25 + \beta^2 = 1$, so $\beta^2 = 0.75$, $\beta = \pm\sqrt{3} / 2$

$\beta = \pm (\sqrt{3})/2$.

Find α if A is perpendicular to B



$A = 2\hat{i} + 3\hat{j} + \alpha\hat{k}$ and $B = \hat{i} - 2\hat{j} + 4\hat{k}$. If A is perpendicular to B, find α.

Solution: Find α if A is perpendicular to B

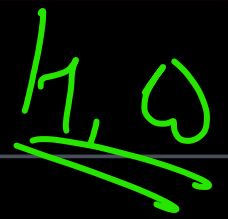
Step 1: $A \cdot B = 0$

Step 2: $2 - 6 + 4\alpha = 0$

Step 3: $4\alpha = 4$, so $\alpha = 1$

$\alpha = 1$.

Angle of a vector from the x-axis



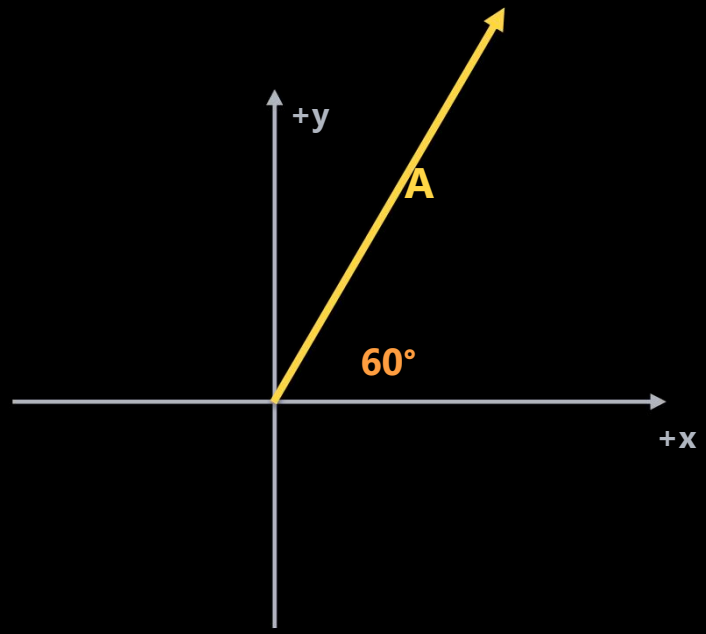
A = 5 î + 5 √3 ĵ. Find the angle of A from the x-axis.

Solution: Angle of a vector from the x-axis

Step 1: $\tan \alpha = A_y / A_x = 5\sqrt{3} / 5 = \sqrt{3}$

Step 2: $\alpha = 60°$

60° from the +x axis.

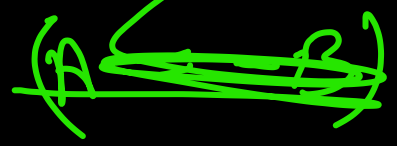


Rough work / extra work

Find the smaller vector

Resultant is perpendicular to the smaller vector and half the larger vector. Find the smaller magnitude in terms of the larger B.

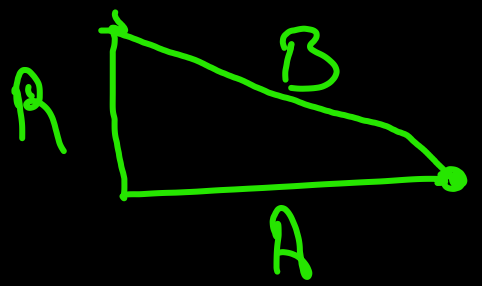
$$(A+B)\cdot A = 0$$



$$A + B = \frac{B}{2}$$

$|A|$ = in terms of B

$$R = \frac{B}{2}$$



$$B^2 = \frac{B^2}{2^2} + A^2$$

$$A^2 = B^2 - \frac{B^2}{4} \Rightarrow \frac{3B^2}{4}$$

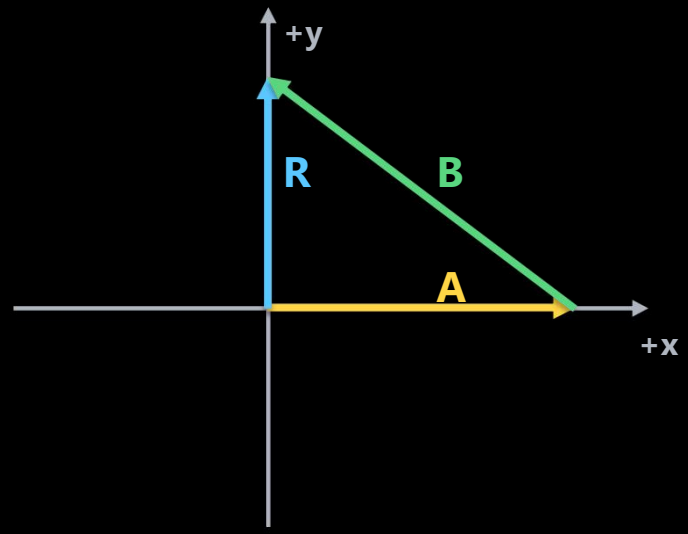
$$\boxed{|A| = \frac{\sqrt{3}}{2} B}$$

Solution: Find the smaller vector

Step 1: Right triangle: $A^2 + (B/2)^2 = B^2$

Step 2: $A^2 = B^2 - B^2/4 = 3B^2/4$

$A = (\sqrt{3} / 2)\ B.$



Three equal vectors OP, OQ, OR

H.W

Three vectors each of magnitude A act as OP along +x, OQ along +y, and OR 45° below +x. The resultant is A √(x). Find x.

Solution: Three equal vectors OP, OQ, OR

Step 1: OP + OQ = $A\sqrt{2}$ at 45° above +x

Step 2: OR is 45° below +x, so that intermediate resultant and OR are at 90°

Step 3: $R = \sqrt{((A\sqrt{2})^2 + A^2)} = A\sqrt{3}$

x = 3.

Rough work / extra work

Vector at 30° from the y-axis

H.W

A vector in the x-y plane makes 30° with the y-axis. Its y-component is 2 √3. Find the x-component.

Solution: Vector at 30° from the y-axis

Step 1: $A_y = A \cos 30° = 2\sqrt{3}$

Step 2: $A(\sqrt{3}/2) = 2\sqrt{3}$, so $A = 4$

Step 3: $A_x = A \sin 30° = 4 \times 1/2 = 2$

x-component = 2.